technical specification

# NOVATION® 3600 34121304

NOVATION® 3600 functional native starch is a tapioca based starch that is recommended for moderate temperature and shear food processing systems. It is ideally suited for use in pasteurized puddings, desserts, soups, sauces, ice cream and other dairy applications, as well as in fruit preparations. This product is produced under Ingredion Incorporated's TRUETRACE® Identity Preserved Program for non-GM products.

## Chemical and Physical Properties

| | Min. | Max. |
|---|---|---|
| Moisture, % | - | 15.0 |
| pH (20% w/w slurry) | 5.0 | 6.5 |
| Viscosity (CML-M106P2) | | |
| End, MVU | 400 | 680 |

## Physical Appearance

| | Typical |
|---|---|
| Color | Off-White to Cream |
| Form | Fine Powder |

## Screen Test

| | Typical |
|---|---|
| % thru U.S.S. #100 | >95 |
| % thru U.S.S. #200 | >85 |

## Microbiological Limits

Initial testing is done on a single composite sample against a limit of m. If result is above m, the three class sampling and acceptance below is used.

| | n | c | m | M |
|---|---|---|---|---|
| Total Plate Count/g | 5 | 3 | 1,000 | 10,000 |
| Yeast/g | 5 | 3 | 200 | 1,000 |
| Mold/g | 5 | 3 | 200 | 1,000 |
| Enterobacteriaceae | 5 | 3 | 10 | 100 |

Where n = # of samples tested; c = maximum allowable number of results between m and M; m = upper target limit; M = maximum acceptable value.

| | |
|---|---|
| *E. coli* | Negative |
| *Salmonella* | Negative |

## Nutritional Data/100 g

| | Typical |
|---|---|
| Calories | 359 |
| Calories from fat | 0 |
| Total Fat, g | <0.1* |
| Cholesterol, mg | 0 |
| Sodium, mg | 34 |
| Total Carbohydrate, g | 89.7 |
| Dietary Fiber, g | 0 |
| Total Sugars, g | 0 |
| Added Sugars, g | 0 |
| Other Carbohydrate, g | 89.7 |
| Protein, g | <0.1* |
| Vitamin D, mcg | 0 |
| Calcium mg | 41 |
| Iron, mg | 0.2 |
| Potassium, mg | <20* |
| Ash, g | <0.1* |

* Not present at level of quantification.

## Certification

Kosher pareve
Halal

## Packaging and Storage

NOVATION® 3600 functional native starch is packaged in multi ply kraft paper bags with a net weight of 55 lbs. NOVATION® 3600 functional native starch should be stored in a clean, dry area at ambient temperature and away from heavily aromatic material.

## Shelf Life

The best before date for NOVATION® 3600 functional native starch is 24 months from the date of manufacture.

## Regulatory Data

| | |
|---|---|
| Source | Tapioca |
| CAS No. | 9005-25-8 |

### United States

Meets FCC (Food Chemical Codex) requirements.

| | |
|---|---|
| Labeling | Tapioca Starch |

### Canada

| | |
|---|---|
| CFDA Regulation | Unstandardized Food |
| Labeling | Tapioca Starch |

## Features and Benefits

NOVATION® 3600 functional native starch has good process tolerance. It performs in food processes where traditional native starches cannot be used as a viscosifier or stabilizer. As a tapioca based product, NOVATION® 3600 functional native starch features a lower hot viscosity which can be an advantage when using processing equipment such as plate heat exchangers. NOVATION® 3600 functional native starch is compatible in the same food processes where modified starches are currently used. No special preparation or equipment is needed to use this starch. When NOVATION® 3600 functional native starch is cooked, it imparts a smooth short texture that sets to a tender gel upon cooling. Because the textural properties closely resemble some modified tapioca starches, only minor adjustments in formulation are necessary to replace modified tapioca starches in food systems. NOVATION® 3600 functional native starch features the inherent bland flavor of tapioca starches plus the unique flavor release characteristic of NOVATION® functional native starches. This product is produced under Ingredion Incorporated's TRUETRACE® Identity Preserved Program for non-GM products.

Effective Date: May 3, 2017 Next Review Date: May 3, 2020

The information described above is offered solely for your consideration, investigation, and independent verification. It is up to you to decide whether and how to use this information. Ingredion Incorporated and the Ingredion group of companies make no warranty about the accuracy or completeness of the information contained above or the suitability of any of their products for your specific intended use. Furthermore, all express or implied warranties of noninfringement, merchantability, or fitness for a particular purpose are hereby disclaimed. Ingredion Incorporated and the Ingredion group of companies assume no responsibility for any liability or damages arising out of or relating to any of the foregoing.

The INGREDION mark and logo are trademarks of the Ingredion group of companies. All rights reserved. All contents copyright © 2017.

5 Westbrook Corporate Ctr.
Westchester, Illinois 60154
U.S.A.
708.551.2600

1600 – 90 Burnhamthorpe Rd., West
Mississauga, Ontario L5B 0H9
Canada
905.281.7950

www.ingredion.us